Love of Variety Simulation

Tate Mason

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1 Model

The consumer's problem is to maximize utility subject to their choice set. The utility function is as follows:

$$U_{ijt} = \beta_i \cdot X_{jt} - \gamma_{ijt}(X_{jt} - \bar{X}_{jt})^2 + \epsilon_{ijt} \tag{1}$$

Where U_{ijt} is the utility of consumer i for product j at time t, β_i is the consumer's sensitivity to the product's attributes, X_{jt} is the attribute of product j at time t, $\gamma_i jt$ is the consumer's sensitivity to sameness of the product's attribute from its average, $\bar{X}_{jt}$ is the average attribute of product j at time t, and ϵ_{ijt} is a random error term distributed T1EV.

The consumer chooses the product that maximizes their utility. The choice set includes all products available in the market at time t. The probability of consumer i choosing product j at time t can be expressed as:

$$P_{ijt} = \frac{e^{U_{ijt}}}{\sum_k e^{U_{ikt}}} \tag{2}$$

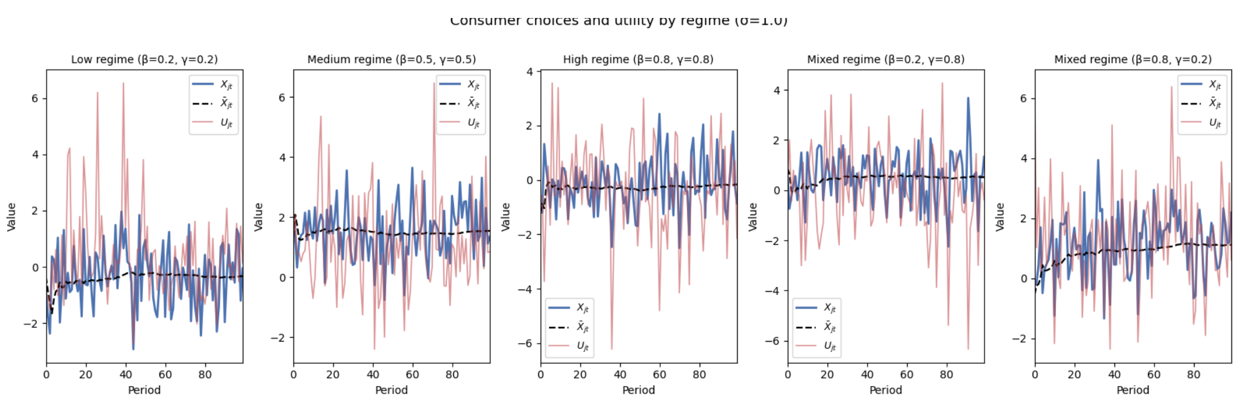
Where the denominator sums over all products k in the choice set at time t.

The simulation will involve setting β and γ at differing levels to observe how consumer choices change.

Further, we will look at the basics of the firm's decision of what to present to the consumer. The firm will "send" three products each period. Define $C_t = \{c_1, c_2, c_3\}$ as the set of products the firm sends at time t. $C_t \sim N(\bar{X_{j,t,x}})$ where $\bar{X_{j,t}}$ is the average attribute of product j at time t and σ_x is the standard deviation of the attributes of the products sent by the firm. We want to see how the firm will react to various regimes in terms of σ_x and how that will affect consumer choices.

2 Results

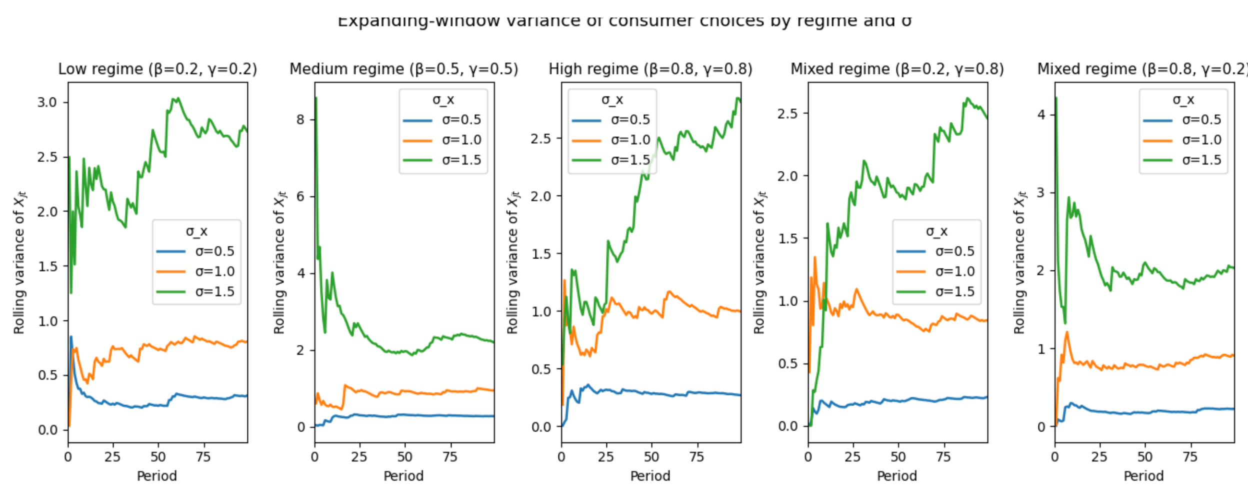
2.1 Consumer Choices and Utility



The blue line represents the chosen product attribute at time t. The orange line represents the utility from that period. The black dashed line is the evolution of the mean attribute. From the figures, we can see that in the low regime, it seems that after about 50 periods, we converge to a "steady state" where utiility is in line with the chosen attribute, implying in this case they have been able to find the "sweet spot". In the medium regime, It takes closer to 75 periods, with a lot of volatility even until the 70th or so period. In this period the "sweet spot" is less clear, as there is still lower utility than attribute value. In the high regime, we actually see a decent fit, but with spikes intermittently. This is likely because when the consumer is more sensitive to variety, the deviations will be more pronounced. They also like what they like more, so this creates an interesting dynamic where they want newness (which they may like), but if it is too new they'll be punished. In the regime with low utility from characteristic and a higher variety parameter, we cannot get a good fit, with utility going all over the place, especially in a negative direction. This is likely because the consumer is very sensitive to newness, and any new draw of σ_x that is too high will cause a sharp drop in utility. In the inverse regime, we see almost the exact opposite, though with an admittedly better fit to the X_{jt} chosen. Here, this is likely because the consumer is just happy to have new product characteristics, and is relatively insensitive to redundancy in product characteristics. This is why we see positive spikes in utility.

All in all, we see that utility is more volatile as X_{jt} deviates from $\bar{X}_{jt}$, a data fact we should expect. We also see that the initial parameters of the consumer have a large effect of the model.

2.2 Rolling Variance of Consumer Choices and σ_x



Now, we can discuss the rolling variance of σ_x. I set it to 3 starting values $\sigma_x = [0.5, 1.0, 1.5]$. Starting in the low regime, the high value of σ_x grows in variance over the entire time horizon while the middle and low value reach steady state after about 40-50 periods. In the medium regime, they converge similarly, with $\sigma_x = 1.5$ having decreasing variance for the first 10 or so periods. The middle and low values do not really evolve, staying fairly constant over the time horizon. In the high regime, All three values grow in variance, but the high value grows the entire time while the other two converge after about 25-35 periods. In the regime with $\beta = 0.2, \gamma = 0.8$, the high value of the high value grows the entire time, while the other two almost immediately converge to a steady state.

In the inverse regime, the high variance falls immediately, before climbing back for about 10 periods, then falling for 20 periods and then converging. The other two converge after about 10 periods.

Here, it seems that without high variance, sigma is not super volatile regardless of regime. When variance is high, however, it is very volatile with respect to γ especially. This is likely because when the consumer is more sensitive to variety, deviations will be more pronounced and so the variance of σ_x will be more volatile.

3 Code

```
import pandas as pd
import numpy as np
import scipy as sp
import matplotlib.pyplot as plt
import seaborn as sns

"""
This file will be used to demonstrate the initial simulation of the consumer's problem
    over 100 periods. We will use one consumer to start, with a pre-determined
set of parameters:
    - beta_low = 0.2
    - beta_high = 0.8
    - beta = 0.5
    - gamma_low = 0.2
    - gamma_high = 0.8
    - gamma = 0.5
    - X_jt ~ N(x_bar, sigma_x^2) where x_bar is drawn from the mean of past choices
    and sigma_x is a parameter that we will want to see react to different regimes
We will simulate the consumer's choices over 100 periods, and then analyze how the
    objective changes over time, how the variance in choices change, and how the
variance changes with regimes. We also want to see how X and sigma comingle.

    Our objective function is:
    U_jt = beta_i*X_jt - gamma*(X_jt - x_bar)^2 + epsilon_ijt
    where epsilon_ijt ~ T1EV
"""

rng = np.random.default_rng(seed=219) # set seed for reproducibility

# Parameters
T = 100
sigma_values = [0.5, 1.0, 1.5] # different variance levels

# Regimes
regimes = [
    (0.2, 0.2, 'Low regime (  =0.2,   =0.2)'),
    (0.5, 0.5, 'Medium regime (  =0.5,   =0.5)'),
```

```
    (0.8, 0.8, 'High␣regime␣(  =0.8,␣  =0.8)'),
    (0.2, 0.8, 'Mixed␣regime␣(  =0.2,␣  =0.8)'),
    (0.8, 0.2, 'Mixed␣regime␣(  =0.8,␣  =0.2)')
]

def simulate(beta, gamma, sigma, T, rng):
    x_choices = np.zeros(T)
    x_bar = np.zeros(T)

    x_bar[0] = rng.standard_normal()
    x_choices[0] = rng.normal(x_bar[0], sigma)

    for t in range(1, T):
        x_bar[t] = np.mean(x_choices[:t]) # mean of all prior choices
        x_choices[t] = rng.normal(x_bar[t], sigma)

    return x_choices, x_bar


def rolling_var(x, min_periods=2):
    var = np.full(len(x), np.nan)
    for t in range(min_periods, len(x) + 1):
        var[t-1] = np.var(x[:t], ddof=1)
    return var

fig, axes = plt.subplots(1, 5, figsize=(13, 5), sharey=False)
idx = np.arange(T)
colors = ["#1f77b4", "#ff7f0e", "#2ca02c"]  # one per sigma value

for ax, (beta, gamma, label) in zip(axes, regimes):
    for sigma, color in zip(sigma_values, colors):
        x_choices, _ = simulate(beta, gamma, sigma, T, rng)
        roll_var = rolling_var(x_choices)
        ax.plot(idx, roll_var, label=f"  ={sigma}", color=color, linewidth=1.8)

    ax.set_title(label, fontsize=11)
    ax.set_xlabel("Period")
    ax.set_ylabel("Rolling␣variance␣of␣$X_{jt}$")
    ax.legend(title=" _x ")
    ax.set_xlim(0, T - 1)

fig.suptitle("Expanding-window␣variance␣of␣consumer␣choices␣by␣regime␣and␣  ",
    fontsize=13, y=1.01)
plt.tight_layout()
plt.savefig("../Output/rolling_variance_of_consumer_choices_by_regime_and_sigma.png")
plt.show()

# Consumer Choices
```

```python
C_x = "#4C72B0"
C_Xb = "black"
C_U = "#C44E52"

fig2, axes2 = plt.subplots(1, 5, figsize=(16, 5), sharey=False)

for ax, (beta, gamma, label) in zip(axes2, regimes):
    x_choices, x_bar = simulate(beta, gamma, sigma_values[1], T, rng)  # using medium sigma for choice simulation

    epsilon = rng.gumbel(size=T)
    U = beta * x_choices - gamma * (x_choices - x_bar) ** 2 + epsilon

    ax.plot(idx, x_choices, label="$X_{jt}$", color=C_x, linewidth=2.0)
    ax.plot(idx, x_bar, label="$\\bar{X}_{jt}$", color=C_Xb, linewidth=1.6, linestyle='--')
    ax.plot(idx, U, label="$U_{jt}$", color=C_U, linewidth=1.2, alpha=0.6)
    ax.set_title(label, fontsize=10)
    ax.set_xlabel("Period")
    ax.set_ylabel("Value")
    ax.legend(fontsize=9)
    ax.set_xlim(0, T - 1)

fig2.suptitle("Consumer␣choices␣and␣utility␣by␣regime␣(  =1.0)", fontsize=13, y=1.01)
plt.tight_layout()
plt.savefig("../Output/consumer_choices_and_utility_by_regime.png")
plt.show()
```